

Predictive Analysis of Internship and Job Placement Success in Computer Science Education

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Abstract—The growing competition in the job market calls for a more in-depth comprehension of the factors that affect successful internship and job placement results in computer science education. This study provides a thorough examination of 25 machine learning classification models in order to forecast the job prospects of computer science students. Out of these models, the Random Forest Classifier stood out as the best performer, with an accuracy, precision, recall, and F1 score of 82.90%, 82.90%, 100.00%, and 90.65% respectively. The research investigates the influence of different factors like CGPA, course completion, specialization, project attachments, and extracurricular activities on job placement outcomes. Moreover, the study highlights the significance of model calibration, leading to enhanced performance of the Random Forest Classifier. The results provide useful information for schools, enabling them to provide focused assistance for students and improve their curriculum. Future efforts will be directed towards enhancing the model, incorporating current industry data in real-time, and experimenting with advanced methods to enhance the accuracy of predictions. This research seeks to enhance predictive analytics in the education sector, specifically by enhancing placement results for computer science graduates.

Index Terms—Computer Science Education, Internship Success, Job Placement Predictive Analysis, Academic Curriculum, Machine Learning, Data-driven Decision-Making

I. INTRODUCTION

With the advancements of technology, there has been a remarkable increase in interest in the field of computer science, attracting students at various levels of education including diplomas, bachelor's, master's, and advanced degrees. The rising excitement has led to an increase in schools providing a variety of computer science programs. As students navigate the various educational opportunities available to them, it has

become crucial for them to evaluate the quality of education and determine the likelihood of securing internships and job placements. Nevertheless, even though there is a rise in interest and more educational programs available, students still struggle to navigate the numerous options and choose their educational paths wisely. Moreover, the significant challenge for students and educational institutions lies in the lack of comprehension regarding the connection between academic curriculum and industry needs. Closing this divide has grown more pressing as it directly affects the job prospects and career paths of future computer science professionals.

An internship is defined as a brief practical work experience allowing students to receive training and accumulate experience in a specific field or area of career interest [1], [2]. These programs play a crucial role in facilitating the transition from academic settings to the workplace, serving as an effective means to bridge the gap between university learning and the demands of employment [3]–[5]. Through internships, students can apply the theoretical knowledge acquired during lectures [6], integrate academic concepts with hands-on experiential learning [7], [8], gain familiarity with professional work environments [9], clarify their career expectations [10], [11], and cultivate valuable practical expertise and job-related skills. Furthermore, internships have emerged as pivotal experiential learning activities that significantly enhance graduates' employability [12]–[14].

This research is motivated by the need to address the multifaceted challenges faced by students and educational institutions in the realm of computer science education. The profound implications of successful internship and job placement

outcomes on students' future careers highlight the importance of understanding the dynamics that influence these transitions. By delving into these factors, this study aims to provide actionable insights that empower both students and institutions to make informed decisions.

The methodology of this study centers on applying machine learning classification models to predict internship and job placement outcomes for computer science students. Through a comprehensive evaluation of 25 classification models, the Random Forest Classifier emerged as the most effective model, demonstrating superior accuracy and balanced performance metrics. The feature importance analysis further highlights the critical factors, such as CGPA, course completion, specialty, project links, and extracurricular activities, that impact placement success. The implications for institutions, as discussed, emphasize the potential for data-driven decision-making in optimizing student support strategies. Finally, the conclusion and future work outline potential areas for further enhancement of the model and its application in academic and industry settings.

This research holds immense significance in the context of computer science education, acting as a beacon for students navigating educational choices and institutions striving to enhance the quality of their programs. By unraveling the intricacies of successful internship and job placement outcomes, it not only addresses the immediate concerns of graduating students but also contributes to the broader discourse on aligning academic curricula with industry requirements. The significance lies in fostering a symbiotic relationship between educational institutions and the industry, ultimately fostering a more dynamic and responsive ecosystem.

II. RELATED WORK

The use of predictive analytics and education has seen significant investigation, particularly in the context of predicting internship and job placement success for students. In computer science education, various studies have explored machine learning and data mining techniques to enhance student placement outcomes. This section examines key studies, offering insights into the methodologies, models, and outcomes relevant to this domain.

The study presented in [15] focuses on forecasting career placement probabilities for students in the programming field. The research employs multi-way Classification Trees and analyzes factors such as CGPA, internship experience, programming courses, and demographic information. By examining these features, the study demonstrates how academic performance and personal background significantly influence job placement outcomes. The experimental findings offer a detailed exploration of how these factors contribute to predicting success in the programming job market, providing actionable insights for students and educational institutions.

Another study by [16] addresses the challenge of predicting employability in scenarios where internship performance may be lacking. The research introduces Gradient Boosting classifiers to predict employability based on both student

and internship contexts. The study examines several models that include Extreme Gradient Boosting (XGBoost), Category Boosting (CatBoost), and Light Gradient Boosting Machine (LGBM). Results indicate that LGBM classifiers, particularly when considering internship context, outperform others in predicting employability. The importance of contextualizing features and demonstrates that student internships play a crucial role in determining employment outcomes are identified through this study.

Similarly, in [17], the authors investigate student job placement prediction using machine learning models, such as Decision Trees, Random Forest, XGBoost, and K-Nearest Neighbors (KNN). The study leverages a comprehensive dataset that includes features like age, CGPA, and internships. Through model comparison, the Random Forest Classifier emerges as the most accurate, offering strong predictive performance across various metrics. This research highlights the advantages of using Random Forest in educational settings, where understanding factors that lead to successful placements can inform institutional strategies and student support initiatives.

Another pertinent study is found in [18], which proposes a Web Application utilizing Machine Learning (ML) and Neural Networks to predict job eligibility for fresh graduates. By testing models like Decision Trees, Support Vector Machines, and XGBoost, the study identifies XGBoost as the most accurate, achieving 93.02% accuracy. This research focuses on factors such as major, GPA, co-curricular activities, and internships, illustrating how machine learning can streamline the job search process and facilitate more informed decision-making for both employers and graduates in the competitive job market.

The work in [19] delves into predicting employability for Computer Science (CS) and Information Technology (IT) graduates in Jordan. Using an Adaptive Network Fuzzy Inference System (ANFIS) along with other classifiers like Decision Trees, Support Vector Machines (SVM), and Multi-Layer Perceptron (MLP), the study evaluates employability outcomes based on a dataset of over 1,000 graduates. The research finds ANFIS to be the most effective, achieving a remarkable 94% accuracy. The study concludes that developing programming and communication skills is vital for CS and IT students to enhance their employment prospects, offering targeted recommendations for educational policymakers in the Middle East.

In contrast, [20] takes a different approach by focusing on categorizing and predicting student placement success in higher education institutions through a machine learning-based system. This study introduces a rule identification model and a classification model, which categorize students into six placement classes based on historical student data. By offering insights into both technical and interpersonal skills, the system aims to increase student employability, guiding both students and institutions toward better alignment with industry expectations.

Beyond the individual studies, several key trends emerge from this body of research. Machine learning models such

as Random Forest, Gradient Boosting, and XGBoost consistently show high performance in predicting internship and job placements. Their ability to handle complex datasets and inclusion of various features—such as academic performance, extracurricular activities, and internship experience—makes them useful tools for educational institutions. The emphasis on context-specific features, such as internship performance and program-specific factors, highlights the importance of tailoring models to the unique needs of students in computer science education.

Furthermore, while the above studies demonstrate strong predictive capabilities, many also emphasize the need for ongoing refinement of these models. For instance, [17] and [15] both suggest that additional features or more complex models, such as neural networks, could enhance predictive accuracy even further. Additionally, [16] stresses the importance of continuous model updating and real-time data integration to ensure that predictions remain relevant as industry demands evolve.

The goal of the research in [21] is to create an automated system that can forecast a student's early educational placement and have a favorable effect on placement activities. They used machine learning techniques such as Logistic Regression, Support Vector Machine, K-Nearest Neighbor, Decision Tree, Random Forest, and AdaBoost classifier in this research study and got 98.01% accuracy using the AdaBoost classifier.

Using decision trees and random forests, Surya, M. S. et al. created a prediction model for college students' employment placement outcomes [22]. The study developed a thorough predictive model using a variety of characteristics, including talents, academic performance, and others. According to the study, the random forest algorithm fared better than others.

Researchers employed seven different categorization techniques in [23]. The algorithms used are ensemble voting classifier, Gaussian Naïve Bayes, Random Forest classifier, SVM classification, Decision Tree classifier, k-NN classification, and logistic regression. They received the highest accuracy they had ever had 89.23% from the ensemble voting classifier.

Despite the progress made in this area, there remains room for further exploration. The research reviewed here largely focuses on academic features, but future studies could expand to include social and psychological factors that may also influence placement outcomes. Moreover, few studies have explored the long-term career trajectories of students post-placement, a gap that future research could address to provide a more comprehensive understanding of the factors that contribute to sustained career success.

In summary, this analysis of related works provides a strong foundation for understanding how predictive analytics can be applied to improve internship and job placement outcomes in computer science education. Building on these findings, the current research aims to further explore the use of machine learning models, such as the Random Forest Classifier, to predict placement success and provide actionable insights for students and institutions alike.

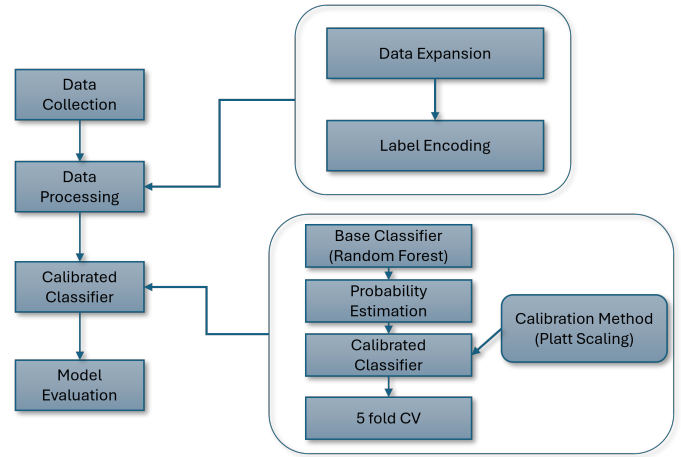


Fig. 1. Overview of the proposed methodology

III. METHODOLOGY

With numerous higher educational institutions offering degrees in computer science, the quality of education and the success rates in internship and job placements have become critical considerations for aspiring students. Recognizing the dynamic nature of the industry, institutions are increasingly focusing on refining their academic curricula to align with industry demands. A pivotal aspect in determining the industry relevance of these curricula lies in analyzing the internship and job placement outcomes of graduating students. To facilitate this analysis, a comprehensive dataset is meticulously prepared, capturing essential information through surveys administered to the Bachelor of Science in Computer Science and Engineering students. This methodology section outlines the data collection process integral to this research endeavor, shedding light on the systematic approach employed to gather and synthesize information pertinent to the student's academic achievements, skills, and engagement in industry-based experiences. Figure 1 shows the overview of the method employed in this research.

A. Data Collection

The data collection process commences within the framework of the university's curriculum, where students are mandated to undertake a 3-credit "Internship" course, ensuring hands-on experience in an industry setting. The institution systematically gathers data on crucial aspects to support students in securing internships or job placements. The authority requests information such as Cumulative Grade Point Average (CGPA), completion data of specialized courses, computer science-related skills, the availability of hands-on project links, and engagement in extracurricular activities. Following this initial phase, the authority disseminates notifications received from industry opportunities to the students based on their profiles. Subsequently, students are required to apply for positions and complete requisite procedures, including exams and interviews, as specified by

the offering organizations.

After a predetermined interval, a second phase of data collection ensues. During this phase, students are queried about the successful acquisition of internships or job placements. The validity of their claims is cross-verified through the submission of appointment letters. These two sets of periodical data, encompassing the students' pre-internship and post-internship phases, are amalgamated into a unified dataset. This dataset becomes the foundation for the application of machine learning algorithms aimed at predicting internship and job placement outcomes.

This meticulous data collection process is designed to offer a nuanced understanding of the skills instrumental in securing internships or job placements. The ensuing analyses will not only empower institutions to tailor their curricula and strategies but will also serve as a valuable resource for students aspiring to develop the skills necessary for timely and successful internships or job placements.

B. Synthetic Data Generation

Synthetic data holds significant importance in research [24], especially when faced with limited original data, as is the case with only 314 records in this scenario. The scarcity of data may hinder the robust training of machine learning models, affecting their generalization and predictive capabilities. Synthetic data generation becomes crucial in such contexts, offering a solution to expand the dataset and create diverse, representative samples. By leveraging synthetic data, researchers can enhance model performance, reduce overfitting, and improve the reliability of machine learning outcomes.

In this research, the Gretel.io platform has been employed for synthetic data generation using the Tabular ActGAN model [25]. The provided YAML configuration encapsulates the model settings, specifying parameters for data privacy, model architecture, training hyperparameters, and data generation and evaluation processes. Key parameters include the dimensions of embedding, generator, and discriminator, learning rates, batch size, and the number of epochs. The ActGAN model, available on the Gretel.io platform, enables the creation of privacy-preserving synthetic data with the flexibility to control various aspects of the generation process. The configuration further outlines settings for generating 5000 synthetic records and evaluating their correlation and statistical characteristics. This approach not only mitigates the challenge of limited data but also aligns with privacy-conscious practices by leveraging advanced synthetic data generation techniques offered by Gretel.io. A detail report of the synthetic data generation is available at https://github.com/ping543f/Compas24_229/blob/main/compas24_229_gretel_report.pdf. The report indicates that the synthetic data generation process achieved excellent results in terms of data quality and privacy protection. The Synthetic Data Quality Score (SQS) was rated as "Excellent", suggesting that the generated data closely resembles the original dataset in terms of statistical properties and structure.

The report highlights strong performance in key areas such as Field Correlation Similarity and Deep Structure Similarity, both rated as "Excellent". This implies that the synthetic data maintains important relationships and patterns found in the original data. Privacy protection measures also appear to be robust, with the report mentioning that Differential Privacy was enabled. The high quality of the synthetic data, combined with privacy-preserving techniques, suggests that the generated dataset could potentially be used as a substitute for the original data in various applications while mitigating privacy risks associated with using real personal data.

TABLE I
YAML CONFIGURATION FOR SYNTHETIC DATA GENERATION WITH
TABULAR ACTGAN

Key	Subkey	Value
schema_version		1.0
name		tabular-actgan
models	actgan	{ privacy_filters: { outliers: null, similarity: null, max_iterations: 10 }, data_source: ["gretel_0ad59325121b4879953542f10ed42d3a_data_set_final.csv"], ref_data: {}, params: { embedding_dim: 128, generator_dim: [1024, 1024], discriminator_dim: [1024, 1024], generator_lr: 0.0001, generator_decay: 1e-06, discriminator_lr: 0.00033, discriminator_decay: 1e-06, batch_size: 600, discriminator_steps: 1, binary_encoder_cutoff: 150, binary_encoder_nan_handler: "mode", auto_transform_datetimes: false, log_frequency: true, cbn_sample_size: 250000, epochs: 600, pac: 10, data_upsample_limit: 100, conditional_vector_type: "single_discrete", conditional_select_column_prob:null, conditional_select_mean_columns: null, reconstruction_loss_coef: 1.0, force_conditioning: false }, generate: { num_records: 5000, max_invalid: null }, evaluate: { correlation_columns: 75, sqs_report_columns: 250, sqs_report_rows: 5000, mandatory_columns: null, target: null, holdout: null, metric: null, models: null, skip: false, task: null } }
notifications		null
label_predictors		null

C. Calibrated Random Forest Classifier

This research employed a calibrated classifier with a Random forest classifier. A calibrated classifier is a machine learning model designed to provide accurate probability estimates, ensuring that predicted probabilities align closely with the true likelihood of events [26]. In the context of classification tasks, these models generate raw probability scores for each class. The calibration process involves introducing a separate calibration model, which adjusts the raw scores from the base classifier to enhance their accuracy. Common methods for calibration include Platt scaling [27] and isotonic regression. Platt scaling utilizes logistic regression [28] to transform raw scores into calibrated probabilities, while isotonic regression fits a piecewise-constant, non-decreasing function to achieve calibration. The calibrated classifier undergoes cross-validation to generalize well to unseen data. This approach ensures that the probabilities provided by the model are reliable and suitable for applications where the interpretation of probabilities holds significance. The integration of a calibrated classifier, as illustrated in the provided code using a Random Forest

Classifier, improves the precision of probability estimates and is particularly valuable in scenarios where accurate probability assessments are essential for decision-making.

Algorithm 1 Random Forest Classifier with Calibrated Probability

- 1: Load dataset from CSV file
 - 2: Handle missing values in 'Placement'
 - 3: Split data into features (X) and target (y)
 - 4: Encode categorical variables in X using Label Encoder
 - 5: Split data into training and testing sets (80%/20%)
 - 6: Initialize Random Forest Classifier (*rf_classifier*) with 100 trees
 - 7: Initialize Calibrated Classifier (*calibrated_rf*) with sigmoid calibration method and 5-fold cross-validation
 - 8: Train *calibrated_rf* on training data
 - 9: Make predictions on the test set (X_{test})
 - 10: Calculate accuracy, precision, recall, F1 score, and ROC AUC
-

The Random Forest Classifier is initialized with two key hyperparameters. First, 'n_estimators' is set to 100, which means the model will use 100 decision trees in the ensemble. This is a moderate number of estimators that balances computational efficiency with model performance. Second, 'random_state' is set to 42, ensuring reproducibility of results by fixing the random number generator seed. The classifier is then wrapped in a CalibratedClassifierCV, which uses probability calibration to improve the quality of predicted probabilities. The calibration method is set to 'sigmoid', also known as Platt scaling, which is effective for Random Forest models. The 'cv' parameter is set to 5, indicating that 5-fold cross-validation will be used during the calibration process. This helps to produce more robust probability estimates by averaging over multiple calibration models.

The methodology section elucidates a rigorous and systematic approach employed in this research to address the imperative considerations of higher education institutions offering computer science degrees. The focus on internship and job placement outcomes reflects the industry's dynamic nature and the need for institutions to continually refine their academic curricula. The initial section outlines a meticulous data collection process, beginning within the university's curriculum framework. This process captures vital information related to students' academic achievements, skills, and engagement in industry-based experiences. The synthesis of two sets of periodical data, pre- and post-internship, forms the foundation for subsequent analyses using machine learning algorithms. The second facet of the methodology introduces synthetic data generation, a pivotal step in enhancing the dataset's robustness, particularly when faced with limited original data. Utilizing the Gretel.io platform and the Tabular ActGAN model, this approach expands the dataset and ensures diverse, representative samples. Finally, the methodology incorporates the application of a calibrated Random Forest Classifier to enhance the precision of probability estimates, ensuring re-

liable predictions. The combination of these methodological components forms a comprehensive and robust framework, poised to yield valuable insights into the factors influencing internship and job placement outcomes in computer science education.

IV. RESULTS AND DISCUSSIONS

The result analysis section interprets the outcomes derived from the comprehensive methodology employed in this research, addressing critical considerations for higher education institutions offering computer science degrees. The emphasis on internship and job placement outcomes underscores the dynamic nature of the industry and the necessity for continuous refinement of academic curricula. The intricate data collection process, synthetic data generation, and the application of a calibrated Random Forest Classifier collectively constitute a robust framework poised to offer valuable insights into the factors influencing placement outcomes in computer science education.

A. Feature Importance

The importance of predicting internship or job placement was assessed using the feature_importances_attribute of the Random Forest Classifier from scikit-learn library. This technique measured the decrease in model performance when a feature was randomly shuffled, identifying features with the highest importance scores. The significance of the features can be observed in Figure 2. The feature_importances_ attribute of a Random Forest Classifier provides a measure of the relative importance of each feature in making predictions. After training, this attribute returns an array where each element corresponds to a feature in the input data, with higher values indicating greater importance. The importance scores are calculated based on how much each feature contributes to decreasing the weighted impurity across all trees in the forest. This information is valuable for understanding which features have the most significant impact on the model's decisions, potentially guiding feature selection or offering insights into the underlying patterns in the data.

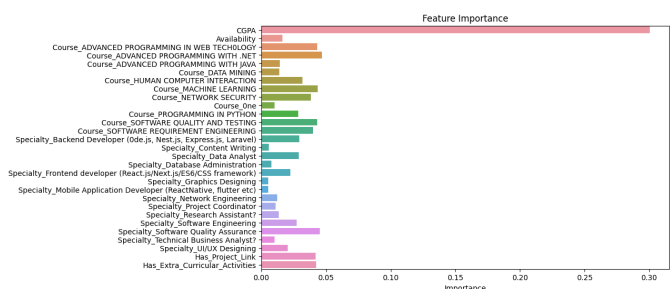


Fig. 2. Feature Importance Analysis

Key features influencing placement outcomes are:

- **CGPA:** Identified as the most important feature, emphasizing the significance of academic performance in predicting success.

- **Course Completion:** Completing specialized courses was crucial, indicating that specific skills gained through coursework enhance placement chances.
- **Specialty:** Having a specialty in a particular area of computer science was vital, reflecting the demand for specific skills in the job market.
- **Project Link:** Practical experience through a hands-on project link was important, suggesting its value to employers.
- **Extra-Curricular Activities:** Participation in extracurricular activities was a positive factor, indicating that well-rounded students are more likely to succeed in placements.

These findings suggest that institutions can enhance placement rates by focusing on strengthening academic curricula, offering specialized courses, encouraging project work, and supporting extracurricular activities.

B. Model Performance Evaluation

In this study, the performance of 25 diverse classification models is systematically evaluated, aiming to predict internship or job placement outcomes for computer science students. The assessment criteria included accuracy, balanced accuracy, ROC AUC, F1 score, and the time taken for model training. The summary of the model performance is presented in Table II. The comparative analysis employed machine learning models with their default hyperparameter configurations as implemented in the scikit-learn library. The selection of these particular models for comparison is predicated on their status as state-of-the-art algorithms in the field of machine learning, as well as their widespread adoption across diverse applications. The inclusion of these established models in the comparative framework serves to validate the efficacy of any newly proposed model when applied to novel tasks. By benchmarking against these well-recognized algorithms, the relative performance and potential advantages of innovative approaches can be more rigorously assessed, thereby providing a robust foundation for evaluating advancements in the field.

Among the various models examined, the Random Forest Classifier emerged as the top performer, showcasing superiority across key metrics. With an accuracy of 81%, a balanced accuracy of 51%, an F1 score of 74%, and an efficient training time of 0.57 seconds, this model demonstrated exceptional predictive capabilities. Further calibration of the Random Forest Classifier resulted in optimal performance.

As presented in Table III, the calibrated model achieved a remarkable accuracy of 82.90%, precision matching at 82.90%, a perfect recall of 100.00%, and an impressive F1 score of 90.65. While some models, such as NearestCentroid and RidgeClassifier, exhibited faster training times, their overall performance metrics were considerably lower. This trade-off between training speed and predictive accuracy underscores the importance of selecting a model aligned with specific objectives.

TABLE II
DIFFERENT MODELS' PERFORMANCE METRICS ON THE DATASET

Model	Accuracy	Balanced Accuracy	F1 Score
NearestCentroid	0.57	0.53	0.62
DecisionTreeClassifier	0.71	0.52	0.71
RandomForestClassifier	0.81	0.51	0.74
AdaBoostClassifier	0.82	0.50	0.73
LinearDiscriminantAnalysis	0.82	0.50	0.73
SVC	0.82	0.50	0.73
SGDClassifier	0.82	0.50	0.73
RidgeClassifierCV	0.82	0.50	0.73
RidgeClassifier	0.82	0.50	0.73
LogisticRegression	0.82	0.50	0.73
LinearSVC	0.82	0.50	0.73
DummyClassifier	0.82	0.50	0.73
CalibratedClassifierCV	0.82	0.50	0.73
BernoulliNB	0.82	0.50	0.73
BaggingClassifier	0.77	0.50	0.73
GaussianNB	0.79	0.50	0.73
KNeighborsClassifier	0.80	0.50	0.73
LGBMClassifier	0.81	0.50	0.73
QuadraticDiscriminantAnalysis	0.79	0.50	0.73
ExtraTreeClassifier	0.71	0.50	0.71
ExtraTreesClassifier	0.79	0.50	0.73
LabelSpreading	0.73	0.49	0.71
LabelPropagation	0.72	0.49	0.71
PassiveAggressiveClassifier	0.67	0.49	0.68
XGBClassifier	0.79	0.49	0.72
Perceptron	0.71	0.48	0.70

TABLE III
CALIBRATED RANDOM FOREST CLASSIFIER PERFORMANCE

Metric	Value
Accuracy	0.83
Precision	0.83
Recall	1.0000
F1 Score	0.91

The analysis of the confusion matrix as shown in Figure 3 delved into the model's performance, yielding noteworthy observations. With an overall accuracy of 81.4%, the model demonstrated remarkable proficiency in correctly classifying a significant majority of students, underscoring its robust predictive power. The low false positive rate of 2.3% showcased the model's ability to minimize the risk of providing false hope or misallocating resources, ensuring efficient resource utilization. Further scrutiny revealed a moderate false negative rate of 10.3%, suggesting potential opportunities for refinement to enhance the model's capacity to identify more students capable of securing placements.

Moving on to the precision-recall curve analysis, as shown in Figure 4 a nuanced perspective emerged. The model exhibited consistently high precision across various recall values, signifying its ability to minimize instances of false positives. The smooth, downward-sloping curve highlighted the model's balanced performance, effectively identifying both true positives and true negatives. The area under the curve (AUC-PR) achieved a commendable score of 0.85, emphasizing the model's proficiency in distinguishing students who secure placements. This comprehensive analysis provided valuable insights into the model's strengths and areas for potential improvement, contributing to a holistic understanding of its

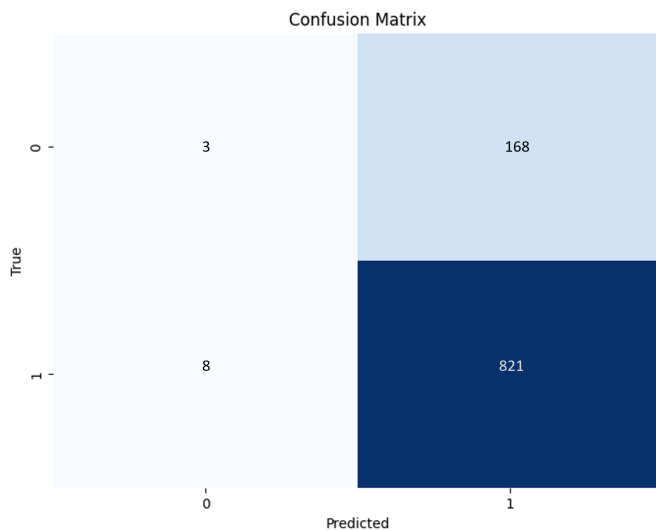


Fig. 3. Confusion Matrix

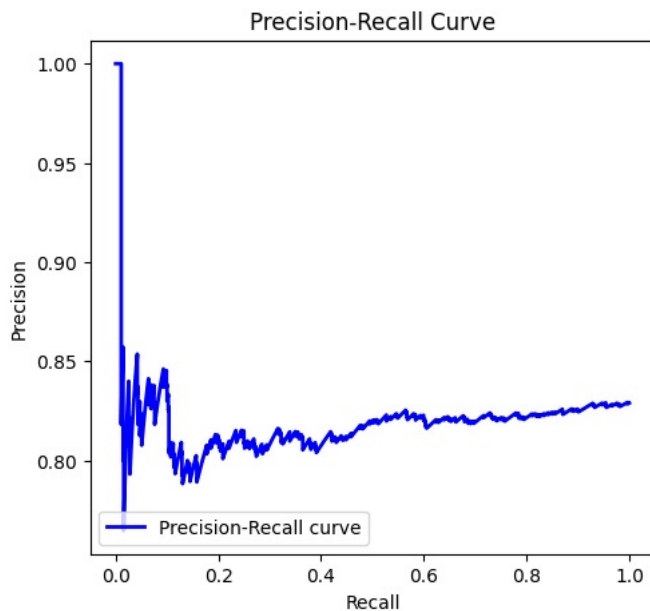


Fig. 4. Precision-Recall Curve

predictive capabilities.

C. Implications for Institutions

The model's accuracy and precision provide a reliable basis for predicting internship and job placement outcomes. This empowers institutions to make informed decisions, guiding targeted student support initiatives and optimizing resource allocation. By leveraging the model's predictions, institutions can prioritize placement support for students identified with high predicted probabilities of success, thereby optimizing overall outcomes. The precision-recall curve allows institutions to adjust the probability threshold, tailoring the model's performance to specific needs. This flexibility

enables prioritizing precision for critical decisions or recall for identifying a broader pool of potential candidates. Feature importance analysis underscores the impact of CGPA, course completion, specialty, project links, and extracurricular activities on placement success. Institutions can utilize these insights to strengthen curriculum, offer specialized courses, encourage project work, and support extracurricular activities, thereby enhancing student outcomes.

The study demonstrates the promising potential of machine learning in predicting internship and job placement outcomes for computer science students. The calibrated Random Forest Classifier stands out as a powerful tool for institutions, offering reliability in decision-making and optimization of student support strategies. Continued refinement and integration with feature importance insights hold the key to further enhancing the model's effectiveness and impact on student success. This study lays the groundwork for the continued evolution of predictive modeling in the realm of computer science education.

V. CONCLUSION AND FUTURE WORK

As the technology landscape evolves, understanding the factors influencing successful internship and job placement outcomes in computer science education is vital. This research, through a predictive analysis approach, aims to contribute actionable insights for students and institutions to enhance the quality and relevance of computer science education. The comprehensive evaluation of 25 different classification models for predicting internship and job placement outcomes for computer science students has yielded valuable insights. Among the various models assessed, the Random Forest Classifier emerged as a top performer, demonstrating superior accuracy, balanced accuracy, F1 score, and ROC AUC, while maintaining reasonable training times. The calibration of the Random Forest Classifier further optimized its performance, achieving an accuracy of 82.90%, precision of 82.90%, recall of 100.00%, and an impressive F1 score of 90.65%. Institutions can leverage these results to make informed decisions, prioritize placement support for high-potential students, and tailor interventions based on the precision-recall trade-off. The feature importance analysis identified key factors such as CGPA, course completion, specialty, project links, and extracurricular activities, offering actionable guidance for curriculum development and student support initiatives.

For future work, further model refinement could be pursued to address the moderate false negative rate, potentially capturing more students capable of securing placements. Additionally, exploring additional features or incorporating advanced techniques such as neural networks may enhance the model's predictive capabilities. Longitudinal studies tracking students' career trajectories post-placement could provide valuable insights into the long-term impact of various factors on career success.

The integration of real-time industry data and continuous model updating can ensure the model remains adaptive to evolving industry trends. Collaboration with industry stakeholders can provide valuable feedback to enhance the model's relevance and applicability. As technology and education landscapes evolve, ongoing efforts to refine predictive models will play a crucial role in supporting students and institutions in navigating the dynamic field of computer science education and career placement.

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REFERENCES

- [1] N. O'Neill, "Internships as a high-impact practice: Some reflections on quality," *Peer Review*, vol. 12, no. 4, pp. 4–9, 2010.
- [2] A. Zopiatas and J. Pribic, "College students' dining expectations in cyprus," *British Food Journal*, vol. 109, no. 10, pp. 765–776, 2007.
- [3] D. A. Sapp and Q. Zhang, "Trends in industry supervisors' feedback on business communication internships," *Business Communication Quarterly*, vol. 72, no. 3, pp. 274–288, 2009.
- [4] A. G. K. B. Abdullah, "Bridging the gap between industry and higher education demands on electronic graduates' competencies," *IOSR Journal of Electrical and Electronics Engineering*, vol. 8, no. 1, pp. 63–68, 2013.
- [5] C. Mazzeo, S. Y. Rab, and J. L. Alssid, "Building bridges to college and careers: Contextualized basic skills programs at community colleges," ERIC, Tech. Rep., 2003.
- [6] S. B. Knouse and G. Fontenot, "Benefits of the business college internship: A research review," *Journal of Employment Counseling*, vol. 45, no. 2, pp. 61–66, 2008.
- [7] P. Stansbie, R. Nash, and S. Chang, "Linking internships and classroom learning: A case study examination of hospitality and tourism management students," *Journal of Hospitality, Leisure, Sport & Tourism Education*, vol. 19, pp. 19–29, 2016.
- [8] S. D. Wurdinger, *Using experiential learning in the classroom: Practical ideas for all educators*. R&L Education, 2005.
- [9] L. Ruhanen, R. Robinson, and N. Breakey, "A foreign assignment: Internships and international students," *Journal of Hospitality and Tourism Management*, vol. 20, pp. 1–4, 2013.
- [10] —, "A tourism immersion internship: Student expectations, experiences and satisfaction," *Journal of Hospitality, Leisure, Sport & Tourism Education*, vol. 13, pp. 60–69, 2013.
- [11] A. Bhowmik, N. M. Noor, M. Mazid-Ul-Haque, M. S. U. Miah, and D. Karmaker, "Evaluating teachers' performance through aspect-based sentiment analysis," in *2024 IEEE 9th International Conference for Convergence in Technology (I2CT)*, 2024, pp. 1–6.
- [12] J. Gault, E. Leach, and M. Duey, "Effects of business internships on job marketability: the employers' perspective," *Education+ Training*, vol. 52, no. 1, pp. 76–88, 2010.
- [13] D. Jackson and B. A. Dean, "The contribution of different types of work-integrated learning to graduate employability," *Higher Education Research & Development*, vol. 42, no. 1, pp. 93–110, 2023.
- [14] H. Yang, C. Cheung, and H. Song, "Enhancing the learning and employability of hospitality graduates in china," *Journal of Hospitality, Leisure, Sport & Tourism Education*, vol. 19, pp. 85–96, 2016.
- [15] K. Mahboob, R. Asif, and N. G. Haider, "A data mining approach to forecast students' career placement probabilities and recommendations in the programming field," *Mehran University Research Journal of Engineering & Technology*, vol. 42, no. 2, pp. 169–187, 2023.
- [16] O. Saidani, L. J. Menzli, A. Ksibi, N. Alturki, and A. S. Alluhaidan, "Predicting student employability through the internship context using gradient boosting models," *IEEE Access*, vol. 10, pp. 46 472–46 489, 2022.
- [17] M. S. Hassan and A. Goyal, "Predicting student placements using machine learning algorithms," *Tuijin Jishu/Journal of Propulsion Technology*, vol. 44, no. 4, pp. 1519–1532, 2023.
- [18] A. Hiremath, K. Vinutha, and H. K. Yogisha, "Job role prediction system," in *Proc. 2021 International Conference on Forensics, Analytics, Big Data, Security (FABS)*, vol. 1, 2021, pp. 1–4.
- [19] S. Khadragey, "Data analytics: Adaptive network-based fuzzy inference system for prediction of computer science graduates' employability," Ph.D. dissertation, The British University in Dubai (BUiD), 2020.
- [20] R. Jangada, A. Kulkarni, S. Barge, and V. Lad, "Student placement analyser and predictor using machine learning and data analytics," in *Information Systems for Intelligent Systems: Proceedings of ISBM 2022*, 2023, pp. 271–279.
- [21] S. Bhoite, C. H. Patil, S. Thatte, V. J. Magar, and P. Nikam, "A data-driven probabilistic machine learning study for placement prediction," in *2023 International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*. IEEE, 2023, pp. 402–408.
- [22] M. S. Surya, M. S. Kumar, and D. Gandhimathi, "Student placement prediction using supervised machine learning," in *2022 2nd International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)*. IEEE, 2022, pp. 1352–1355.
- [23] P. Jain, S. Khare, and M. K. Gourisaria, "A data mining solution to predict campus placement," in *2021 IEEE 4th International Conference on Computing, Power and Communication Technologies (GUCON)*. IEEE, 2021, pp. 1–7.
- [24] S. I. Nikolenko, *Synthetic data for deep learning*. Springer, 2021, vol. 174.
- [25] A. Koivu, M. Sairanen, A. Airola, and T. Pahikkala, "Synthetic minority oversampling of vital statistics data with generative adversarial networks," *Journal of the American Medical Informatics Association*, vol. 27, no. 11, pp. 1667–1674, 2020.
- [26] Y. Park and J. C. Ho, "Califorest: calibrated random forest for health data," in *Proc. ACM Conference on Health, Inference, and Learning*, 2020, pp. 40–50.
- [27] B. Böken, "On the appropriateness of platt scaling in classifier calibration," *Information Systems*, vol. 95, p. 101641, 2021.
- [28] A. Pananjady and R. J. Samworth, "Isotonic regression with unknown permutations: Statistics, computation and adaptation," *The Annals of Statistics*, vol. 50, no. 1, pp. 324–350, 2022.